NOVA SCHOOL OF SCIENCE & TECHNOLOGY

DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

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Master/BSc in Name of Previous Degree

MISSILE INTERCEPTION WITH REACHABILITY TOOLS

DRAFT

Dissertation Plan
MASTER IN ELECTRICAL AND COMPUTER ENGINEERING
SPECIALIZATION IN CONTROL

NOVA University Lisbon
Draft: *January 23, 2026*

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Acronyms

CBF Control Barrier Function *(p. 9)*
CLF Control Lyapunov Function *(p. 9)*

GNC Guidance, Navigation and Control *(p. 7)*

LQR Linear-Quadratic Regulator *(p. 8)*

MPC Model-Based Predictive Control *(p. 8)*

NMPC Nonlinear Model-Based Predictive Control *(p. 9)*

PID Propotional-Integral-Derivative Control *(p. 8)*
PN Proportional Navigation *(p. 7)*

QP Quadratic Programming *(p. 9)*

UAVs Unmanned Aerial Vehichles *(p. 6)*

Chapter 1

Introduction

Missile interception has long been a concern, but traditional solutions remain expensive and require infrastructure that may not be available to many countries. However, advancements in quad-rotor automation technologies have enabled new solutions that require little infrastructure and are much cheaper.

The purpose of this project is to design a controller for a quad-rotor such that it defends a critical structure from a rocket heading towards it, whether by intercepting it or deviating the rocket's trajectory from the structure. The strategy considered for the controller design will be based on reachability tools, which utilises the concept of a "reachable set" - the collection of all states a system can attain from a given initial condition under all admissible control inputs over a specified time horizon.

Notation The set of real numbers is denoted by $\mathbb{R}$. Given a vector $\vec{v} \in \mathbb{R}^n$, it's transpose is denoted by $\vec{v}'$, and likewise for a matrix $\mathbf{M}$. The operation diag($\vec{v}$) creates a matrix with diagonal elements v and off-diagonal elements 0. The inner product between two matrices or vectors is denoted by $\vec{v} \cdot \mathbf{M}$. The cross product operation between a vector and a matrix, denoted by $\vec{v} \times \mathbf{M}$, is equivalent to the inner product between the skew asymmetric matrix of that vector and that matrix, which can be denoted by $[\vec{v}] \cdot \mathbf{M}$. The operation vec($\mathbf{M}$) returns the vectorized matrix.

When expressing the coordinates of a point or vector on an object, it may be expressed in the inertial reference frame, denoted by $\mathcal{F}_I$, which refers to the coordinate system whose origin is fixed on the ground, or the body reference frame, denoted by $\mathcal{F}_B$, which has it's origin on the center mass of the object. When a vector has an uppercase B in overscript ($\vec{v}^B$), it's expressed in the body reference frame. In the absence of an overscript notation, assume it's expressed in the inertial reference frame.

1.1 Modeling

Mathematical models are required to describe the flight dynamics of both rocket and drone. This project utilises an ideal point-mass model that considers ideal responses.

1.1.1 Rocket

The rocket's model is built on the following assumptions.

Mass: Remains constant.

Propulsion: The rocket is powered by a two-pulse rocket motor, with lateral pulse jet logic for control.

Aerodynamics: The rocket is considered to move in a vacuum.

The ideal rocket's model is represented by the state vector $x_R \in \mathbb{R}^{18}$:

$$\dot{x}_R = f_R(x_R, u_R) = \begin{bmatrix} f_p(x_R, u_R) \\ f_v(x_R, u_R) \\ f_\omega(x_R, u_R) \\ f_A(x_R, u_R) \end{bmatrix} = \begin{bmatrix} \vec{v}_R \\ \mathbf{A} \cdot \vec{a}^B + \vec{g} \\ \mathbf{J}^{-1} \cdot (\vec{\tau}^B - \vec{\omega}^B \times \mathbf{J} \cdot \vec{\omega}^B) \\ \text{vec}(-\vec{\omega}^B \times \mathbf{A}) \end{bmatrix} = \begin{bmatrix} \dot{\vec{p}}_R \\ \dot{\vec{v}}_R \\ \dot{\vec{\omega}}^B \\ \text{vec}(\dot{\mathbf{A}}) \end{bmatrix}$$

In which:

$\vec{p}_R \in \mathbb{R}^3$ is the position of the center of mass, expressed in $\mathcal{F}_I$.

$\vec{v}_R \in \mathbb{R}^3$ is the linear speed of the center of mass, expressed in $\mathcal{F}_I$.

$\vec{\omega}^B \in \mathbb{R}^3$ is the angular velocity of the rocket, expressed in $\mathcal{F}_B$.

$\mathbf{A} \in \mathbb{R}^{3\times3}$ is Euler's rotational matrix of the rocket, expressed in $\mathcal{F}_I$.

Additional variables are defined as:

$\vec{a}^B \in \mathbb{R}^3$ is the thruster acceleration vector acting on the rocket, expressed in $\mathcal{F}_B$.

$\vec{g} \in \mathbb{R}^3$ is the gravitational acceleration vector, expressed in $\mathcal{F}_I$.

$\mathbf{J} \in \mathbb{R}^{3\times3}$ is the rocket's inertia matrix.

$\vec{\tau}^B \in \mathbb{R}^3$ is the torque vector on the rocket body, expressed in $\mathcal{F}_B$.

Furthermore, $\mathbf{J}$, $\vec{\tau}^B$ and $\vec{a}^B$ are given by the expressions:

$$\mathbf{J} = \text{diag}(\frac{1}{4}m \cdot R^2 + \frac{1}{12}m(2L)^2, \quad \frac{1}{4}m \cdot R^2 + \frac{1}{12}m(2L)^2, \quad \frac{1}{2}m \cdot R^2)$$

$$\vec{\tau}^B = \begin{bmatrix} 0 \\ 0 \\ -L \end{bmatrix} \times (m \cdot \vec{a}^B)$$

$$\vec{a}^B = \begin{bmatrix} a_x^B \\ a_y^B \\ a_z^B \end{bmatrix}$$

Finally, the lateral acceleration input vector $u_R \in \mathbb{R}^2$ is defined as

$$u_R = \begin{bmatrix} a_x^B \\ a_y^B \end{bmatrix}$$

1.1.2 Drone

The drone's model is built on the following assumptions.

Mass: Remains constant.

Propulsion: Considered to be a vector acting on the center of mass, which responds instantly to input actuation.

Aerodynamics: The drone is considered to move in a vacuum.

The drone's dynamics are simplified as a double integrator in three dimensions. They're given by:

$$\dot{x}_D = f_D(x_D, u_D) = \begin{bmatrix} f_p(x_D, u_D) \\ f_v(x_D, u_D) \end{bmatrix} = \begin{bmatrix} \vec{v_D} \\ \vec{a_D} + \vec{g} \end{bmatrix} = \begin{bmatrix} \dot{\vec{p_D}} \\ \dot{\vec{v_D}} \end{bmatrix}$$

In which:

$\vec{p}_D \in \mathbb{R}^3$ is the position of the center of mass, expressed in $\mathcal{F}_I$.

$\vec{v}_D \in \mathbb{R}^3$ is the linear speed of the center of mass, expressed in $\mathcal{F}_I$.

Additional variables are defined as:

$\vec{a}_D \in \mathbb{R}^3$ is the thruster acceleration vector acting on the rocket, expressed in $\mathcal{F}_B$.

$\vec{g} \in \mathbb{R}^3$ is the gravitational acceleration vector, expressed in $\mathcal{F}_I$.

The input vector $u_D \in \mathbb{R}^3$ is defined as

$$u_D = \vec{a}_D$$

1.2 Problem Statement

The scenario in study can be thought of a defense-intercept game with two asymmetric players, with each player having their hierarchy of objectives to satisfy.

The drone's hierarchy of objectives:

- Dynamic target interception.
- Static area defense.

And the missile's:

- Static target interception.
- Dynamic pursuer evasion.

Each objective poses its own control problem, but each player's GNC structure must account for all its objectives. Regarding a player's navigation system, it must be able to estimate its own and the opponent's state relative to each other and relative to the critical area's position. As for the guidance system, it must apply decision-making based on the player's navigation data and it's goals to plan the most optimal trajectory. Finally, the flight control system calculates the appropriate actuation commands given the player's dynamics, based on its navigation and guidance data.

For the flight control system, it is necessary to define a control law. Since either player is described by a non-linear model, several methods may be used, such as model linearization around an ideal trajectory or NMPC. Optimal Control-based methods, which obtain the optimal input actuation by minimizing some cost function related to the objective, stand out for their ability to join guidance with control, as they do not need explicit path planning and trajectory generation to generate the optimal input. These characteristics favor the control problems at hand, seeing that players do not need to follow specific trajectories, only to converge on their targets while satisfying their other objectives. CLF/CBF-type controllers are also well suited to these problems, since the evasion and defense objectives are essentialy barrier conditions.

For the scenario in which the rocket does not evade the drone, let $V_R : \mathcal{X} \to \mathbb{R}$ a quadratic function representing the distance between the rocket and the critical structure.

The rocket's objective can be represented by the optimization problem:

$$\begin{aligned} \min_{u} \quad & V_R(x,u) \\ \text{s.t.:} \quad & \dot{x} = f_R(x,u) \\ & x \in \mathcal{X}_R \\ & u \in \mathcal{U}_R \end{aligned}$$

The drone's objective is dual; to intercept the rocket as well as diverting the rocket away from the critical structure. Let $V_D : \mathcal{X} \to \mathbb{R}$ a quadratic function representing the distance between the drone and the rocket, and $V_{\text{crit}} : \mathcal{Z} \to \mathbb{R}$ a quadratic CBF representing the intersection between the reachable set of the rocket and the critical area. Furthermore, consider $\gamma(\cdot)$ is a $\mathcal{K}$ class function. The drone's objective can be represented by the optimization problem:

$$\begin{aligned} \min_{u} \quad & V_D(x,u) \\ \text{s.t.:} \quad & \dot{x} = f_D(x,u) \\ & \dot{V}_{\text{crit}}(z) \leq -\gamma(V_{crit}(z)) \\ & x \in \mathcal{X}_D \\ & z \in \mathcal{X}_R \\ & u \in \mathcal{U}_D \end{aligned}$$

Chapter 2

State of the Art

The topics relevant to this project include that of modelling Unmanned Aerial Vehichles (UAVs), specifically of missiles and quadrotor drones; control techniques for guided missiles, specifically with stationary targets, and with adversary avoidance; control techniques for quadrotor drones, specifically with non-cooperative intercept targets. Therefore it's necessary to map out the state of the art on these topics.

2.1 Simulation

It would be impractical to iterate the design of such a control function based on real flight tests alone, specially in the early stages of the project, as each change would require a new, lengthy flight test. Therefore it's necessary to simulate the scenario under study, meaning, to simulate the missile and drone dynamics such as the aerodynamic effects acting on the vehicles, in response to input actuations. An almost ubiquitous tool used in aerodynamical simulation and control design are MATLAB and Simulink [**matlab**], thanks to the diversity of applications and efficiency they provide. Moreover, obtaining an accurate mathematical model for the vehicle dynamics is crucial, whether it be for simulation or designing the flight systems.

2.2 Modelling

2.2.1 UAV Dynamics

Regarding the modelling of missile dynamics, the most relevant phases of flight for this study are the midflight and homing phases. The rotational dynamics of the missile may be parametrized with Euler angles or quaternions. [REFERENCE] When it comes to quadrotor drone dynamics, this study is most concerned with the post-launch flight phase. In [**XiadongZ**], it is shown that the full dynamics of a quadrotor are complicated, strongly nonlinear, inherently unstable with coupled system variables and under-actuated.

However, they may be thought of as two distinct control problems; the outer loop tracking the position and the inner loop tracking the attitude, which should simplify the controller design process.

2.2.2 Aerodynamic Effects

Both the missile and the quadrotor are expected to move at very high speeds. Therefore, it's clear that aerodynamic effects such as air drag must be considered in the modelling process. In the case of quadrotors, blade flapping and thrust variations pose a significant disturbance to the system, and must also be accounted for. In [**HaomiaoH**], these effects are explicitly compensated for in the control loop of a quadrotor drone.

2.3 GNC

The systems that control the movements of UAVs can be described by their three core functions: Guidance, Navigation and Control (GNC). Guidance refers to the systems that determine the desired paths from the vehichle's current position to the target waypoint, as well as establishing the changes in velocity, rotation or acceleration to achieve this course. Navigation refers to the systems that determine, at a given instant of time, the vehicle's location, velocity, rotation, and angular rate. These variables are the system's state vector. Control is referring to the manipulation of forces and torques, by means of actuators (i.e., steering device, thrusters, etc.), needed to follow the desired path while maintaining vehicle stability [**PesceV**]. The goal of this project is, essentially, to design a control function that given the current state vector of both the missile and the quadrotor drone, will drive the drone to intercept the missile.

2.4 Control Systems

It's worthwhile to review existing flight control systems, for both missiles and drones. The most widely used methodologies for missile control systems are Proportional Navigation (PN) guidance, and for UAVs they are Optimal Control-based. These may further be classified into Linear Control Systems and Model Based Nonlinear Control Systems [**FaridK**].

2.4.1 Linear Control Methods

Linear control methods are the least computationally complex methods and the easiest to implement, and generally consist in driving the controlled system variable to converge to a desired setpoint, based on negative feedback. However, they tend to have limited robustness when the desired value is variable, or when the vehicle is under-acted, this is, when there are fewer actuators than the number of degrees of freedom to be controlled. To

overcome these limitations, the controlled system must be linearized, usually by finding linear dynamics around a stable operating point or trajectory.

PID One of the most common Linear control methods is the Propotional-Integral-Derivative Control (PID) control. PID is based on negative feedback, and it aims to converge the controlled variable's value to a given setpoint, according to given specifications. From these specifications the proportional, integral, and derivative gain are selected. These will be multiplied by the respective parts of the error signal, meaning the difference between the desired setpoint and the current value of the controlled variable. This transformed signal is injected back into the system as input. In [**CostaA**] a PID controller is sufficient to reliably drive a quadrotor drone to a waypoint.

LQR Another commonly used linear control method is the Linear-Quadratic Regulator (LQR). The system dynamics are described by a set of linear differential equations and the cost is described by a quadratic function, usually defined as the geometric norm of the difference between the state vector and the setpoint at the current instant. Weighting factors are selected then applied to different parts of the cost function, such as the final value or the sum of all deviations. The control actuation may also feature in the cost function. The LQR provides the optimal control law given these parameters and is robust to perturbations. Moreover, it can handle time-varying models. In [**ShahidaK**], a quadrotor drone is controlled with an LQR based controller. It is shown that this controller is robust, with low steady state error.

Gain Scheduling Gain Scheduling is another linear control method, in which non-linear system dynamics are modelled as a collection of simplified linear models, with each model representing a particular operating regime. Each model has a corresponding linear control law. In [**JingQ**], a gain scheduling-based controller is implemented in a quadrotor drone, where different PID controllers are used depending on the current operating regime. The study shows that, relative to the singular PID controller, the performance in trajectory tracking is increased, while maintaining the advantages of PIDs.

2.4.2 Nonlinear Control Methods

Model-Based Nonlinear Controllers are another family of flight control systems. They have the advantage of being able to handle systems' nonlinear dynamics, making them more robust and versatile than linear control methods. This comes at the cost of a more complex, more computationally expensive implementation.

MPC Model-Based Predictive Control (MPC) as a model-based control method is an extension of the LQR. However, the MPC estimates the model output in a finite time horizon then minimizes tracking error for this horizon. Usually, the control law is updated

online. Designing the MPC controller requires an explicit linear model of the system, and this is not usually the case in UAV control, making linearization necessary. Study [**AtharvaG**] considers both a UAV dynamic target interception scenario and a surveillance scenario, and proposes a novel MPC design with variable time steps. The controller was tested both in simulation and real-world flight tests, in 2D and 3D space, and was shown to be effective and efficient.

NMPC Nonlinear Model-Based Predictive Control (NMPC) is a variation of the linear MPC that can handle nonlinear models. The optimization problem is usually non-convex, therefore this method does not usually yield a closed-form expression and requires numerical methods to obtain the optimal control law for this time step. This introduces the "curse of dimensionality", where the complexity grows exponentially with the horizon. Study [**DoraN**] tackles the problem of collision avoidance in a multi-UAV agricultural scenario. It compares the performance of three NMPC-based control strategies: distributed NMPC where the NMPCs of the UAVs are coupled through nonlinear constraints, distributed NMPC where the NMPCs of the UAVs are coupled through penalty costs, and imposing flight corridors by modelling uncertainties and disturbances. It shows that in a scenario with no path intersections the flight corridor strategy ranks better, however it fails to ensure collision avoidance when there are path intersections, while the other two methods are generally effective but with varying performances.

Backstepping Backstepping Control is a nonlinear control method, commonly used in systems with a relative degree higher than one. This method separates the system into various subsystems such that the relative degree of each is one, then a "virtual control law" is applied to the higher-order subsystem such that it stabilizes the lower order one. This creates a series of "virtual controllers" that ultimately allow the controlled variables to converge towards the setpoint. This allows for an iterative, systematic design process that especially favors the layered structure of UAV dynamics. However, the complexity of these controllers rises dramatically as the relative degree increases, since each differentiation step is increasingly complicated to solve analytically. Study [**<empty citation>**] compares the performances of two route-planning approaches for a UAV controller with collision avoidance based on backstepping: a safety ball approach where the UAV stays a minimum distance away from obstacle boundaries while converging towards the target, and a point-mass approach that treats obstacles as a sphere with radius equal to the obstacle's largest dimension. It shows that the safety ball approach has better performance when it comes to computational complexity, but does not minimise the tracking error.

CLF-CBF Control Lyapunov Function (CLF), along with Control Barrier Function (CBF) and unified by Quadratic Programming (QP) establish their own class of nonlinear controllers. The CLF uses the Lyapunov function, which represents an objective. With the CLF, the input norm is minimized while subject to an inequality constraint on the

derivative of a Lyapunov function, and the system is guaranteed to converge exponentially towards the objective. CLF methodology allows the system to have several objectives simply by adding CLFs representing them to the constraints. CBFs are similar to CLFs, except they use Lyapunov-like functions called Barrier functions to define a boundary that the system cannot cross, and will diverge away from [**AmesA**]. By joining CLFs and CBFs, a powerful, a robust controller is achieved. Other methods may be used along with CLF/CBF controllers to further expand their efficacy, such as backstepping. The study [**LiB**] proposes a defense-winning strategy for active target protection involving multiple nonlinear players, utilising a CBF-QP controller.

2.4.3 Reachability-Based Guidance

PN and optimal control based methods, while traditionally used have their limitations. PN guidance cannot account for changes in the adversary's behavior, while optimal control cannot guarantee feasibility given the current conditions. A modern alternative is Reachability based guidance, that aims to grapple with these limitations. Reachability analysis can be used to implicitly guarantee feasibility of a trajectory, or for collision avoidance. Study [**AkalinB**] employs a novel method for determining the reachable set boundaries for aerodynamic interceptors.

Approximate Geometric Method One way to calculate the reachable set is the Approximate Geometric Method. It approximates the reachable set of the model by manipulating convex shapes such as ellipsoids, zonotopes, or polytopes such that they closely fit the bounds of the expected set in the next time steps. This method can even be used with nonlinear models, as shown in [**SilvestreD1**]. A drawback of these methods is that, as the set computation is iterated for each following time step, the dimensional complexity grows exponentially, thus increasing the computational burden of this method. However, study [**RegoF**] proposes an algorithm that significantly reduces this effect.

2000 Missile Interception with Reachability Tools John Doe

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